

Customer Lifetime Value Prediction for Auto Insurance

A Complete Data Science Portfolio Project

Dataset: Vehicle Insurance Customer Data (Kaggle)

Model: Random Forest Regressor | Best R2: 0.9115

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1. Project Title & Overview

This project predicts Customer Lifetime Value (CLV) for auto insurance customers using supervised regression. CLV quantifies the total revenue expected from a single customer over their relationship with the company.

Accurate CLV prediction enables insurance companies to:

- Target retention campaigns at high-value customers
- Optimize premium pricing per risk segment
- Allocate marketing spend more efficiently

ML Problem Type: Regression (continuous target -- CLV in dollars)

Dataset Size: 9,134 rows, 24 columns

Best Model: Tuned Random Forest (R2=0.9115, RMSE=0.198 on log scale)

2. Dataset Description

Source: <https://www.kaggle.com/datasets/ranja7/vehicle-insurance-customer-data>

File: AutoInsurance.csv | Size: 1.59 MB | Rows: 9,134 | Columns: 24

Missing values: None | Duplicates: 0

Feature	Type	Description
Customer Lifetime Value	Float	TARGET -- total CLV in dollars
Customer	String	ID -- dropped before modeling
State	Categorical	Customer state
Response	Categorical	Response to last marketing campaign
Coverage	Categorical	Insurance coverage level
Education	Categorical	Customer education level
Effective To Date	Date	Policy effective date (month extracted)
EmploymentStatus	Categorical	Employment status
Gender	Categorical	Customer gender
Income	Numerical	Annual income (\$)
Location Code	Categorical	Urban / Suburban / Rural
Marital Status	Categorical	Marital status
Monthly Premium Auto	Numerical	Monthly auto premium paid (\$)
Months Since Last Claim	Numerical	Recency of last claim
Months Since Policy Inception	Numerical	Policy age in months
Number of Open Complaints	Numerical	Active complaint count
Number of Policies	Numerical	Number of policies held
Policy Type / Policy	Categorical	Policy type and sub-type
Renew Offer Type	Categorical	Type of renewal offer

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Sales Channel	Categorical	How policy was sold
Total Claim Amount	Numerical	Total historical claims (\$)
Vehicle Class / Size	Categorical	Vehicle classification and size

3. Workflow Diagram

Raw CSV (AutoInsurance.csv, 9134 rows)

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[Phase 0] Project Setup

- Folder structure, src modules, .gitignore, GitHub repo

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[Phase 1] Download Dataset

- Kaggle API -> data/raw/AutoInsurance.csv

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[01_Data_Preparation.ipynb] EDA

- Shape, dtypes, missing values, duplicates
- Target distribution (skewness: 3.99 raw -> 0.05 log)
- Numerical distributions, categorical value counts
- Correlation heatmap, CLV by Coverage / Vehicle Class

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[02_Feature_Engineering.ipynb] Feature Engineering

- Parse Effective To Date -> Effective_Month
- Outlier analysis (IQR) -- kept all (legitimate CLV extremes)
- Log-transform target: $CLV_{log} = \log_{1p}(CLV)$
- 4 interaction features engineered
- One-hot encode 14 categorical columns
- Shape after encoding: 9134 rows x 80+ features

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[03_Feature_Selection.ipynb] Feature Selection

- LASSO (LassoCV) -- non-zero coefficient features
- RFE (GradientBoosting) -- top 30 by recursive elimination
- Consensus = LASSO union RFE + engineered features

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[04_Model_Training.ipynb] Model Training

- 11 models trained and compared
- Best: Random Forest ($R^2=0.9106$)
- Results saved to reports/model_results.csv

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[05_Model_Evaluation.ipynb] Model Evaluation

- GridSearchCV tuning: R^2 improved to 0.9115
- SHAP explainability: Monthly Premium is top predictor
- Residual analysis: mean~0, normally distributed
- Model saved: models/best_random_forest.pkl

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[main.py] Full Pipeline Runner

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reports/ + images/ (committed to GitHub)

4. Phase-by-Phase Explanation

Phase 0: Project Setup

Created a production-quality folder structure with separate directories for raw data, processed data, notebooks, source modules, models, images, reports, and scripts.

Core source modules written:

- src/config.py: all paths and hyperparameters in one place
- src/data_loader.py: load and validate raw CSV
- src/preprocessing.py: imputation, date parsing, feature engineering, encoding
- src/model.py: train, evaluate, tune, and save all models
- src/visualize.py: all plot functions, saving to images/

GitHub repo created via Python urllib (avoids curl JSON issues on Windows).

Verified: .env is gitignored before first push.

Phase 1 (EDA): 01_Data_Preparation.ipynb

Key findings:

- Dataset: 9,134 rows, 24 columns, no missing values, no duplicates
- Target (CLV) range: \$1,898 to \$83,325 (mean \$8,005)
- CLV is heavily right-skewed (skewness=3.99) -- log transform required
- After log transform: skewness drops to 0.05 (near-normal)
- Monthly Premium Auto has the highest positive correlation with CLV
- Extended Coverage customers have the highest median CLV
- Luxury Car and Sports Car vehicle classes show highest CLV
- Employed customers show higher CLV than unemployed

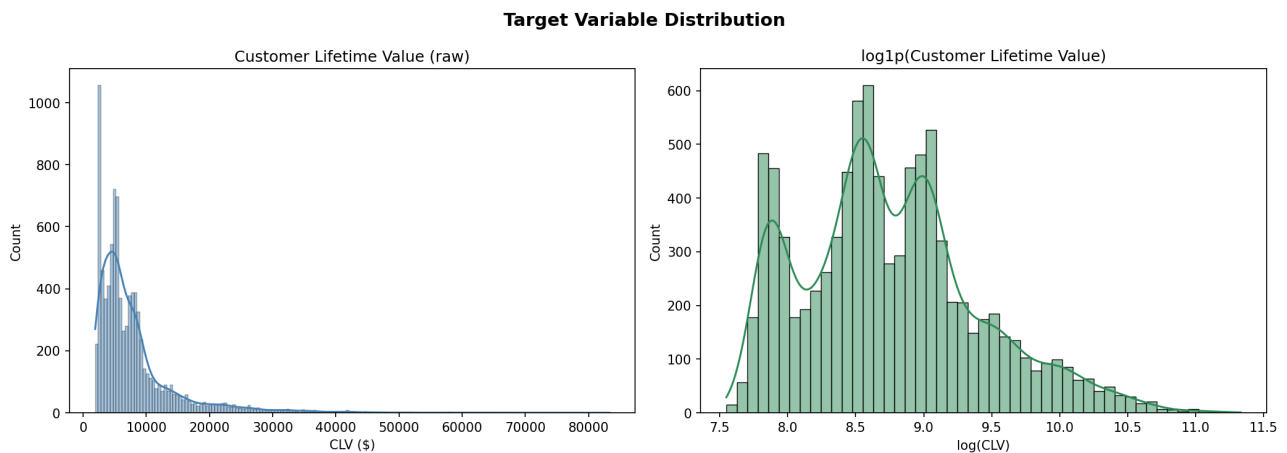


Figure 1: CLV raw vs log-transformed distribution

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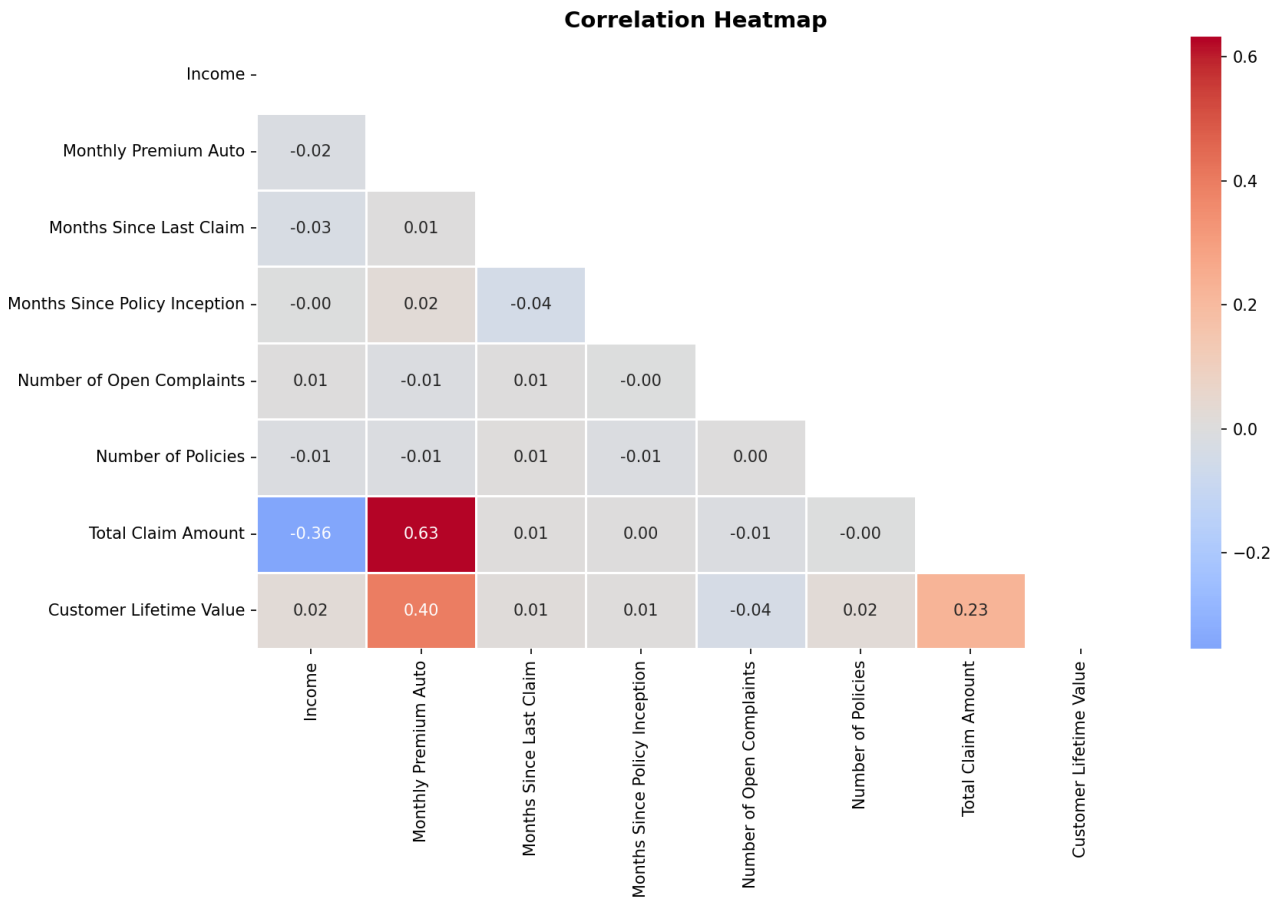


Figure 2: Feature correlation heatmap

Phase 2 (Feature Engineering): 02_Feature_Engineering.ipynb

Decisions made:

- Effective To Date: extracted month number (seasonality signal) and dropped raw date
- Outliers: IQR analysis run on all numerical features. Decision: KEEP all outliers.
Reason: insurance CLV legitimately contains high-value customers; capping would destroy the signal we are trying to predict.
- Log-transform applied to target ($CLV_log = \log_{10}(CLV)$)
- 4 interaction features created (see Feature Engineering table)
- 14 categorical columns one-hot encoded (drop_first=True to avoid multicollinearity)
- Final shape: 9,134 rows x 80+ features

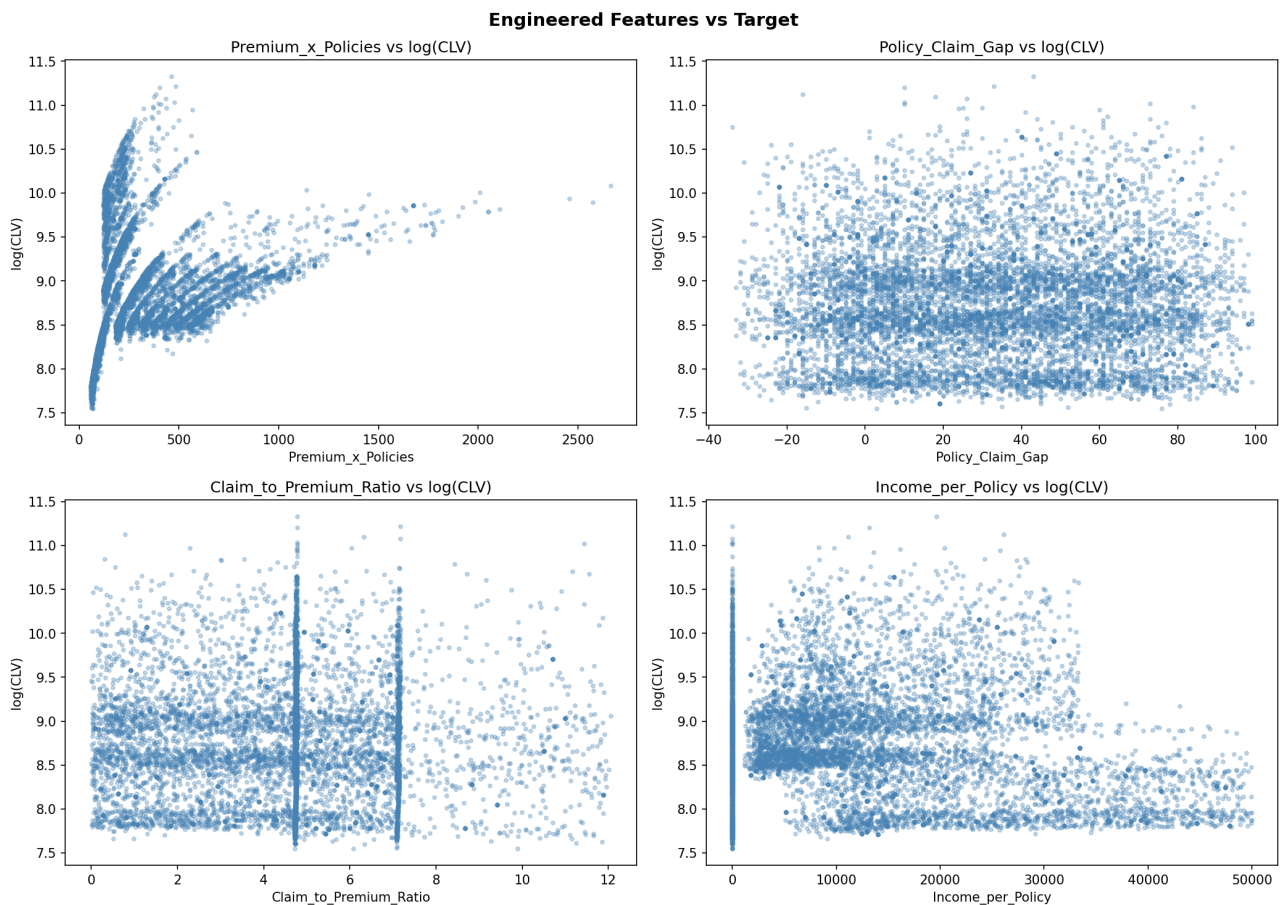


Figure 3: Engineered features vs log(CLV)

Phase 3 (Feature Selection): 03_Feature_Selection.ipynb

Why feature selection: with 80+ features after one-hot encoding, multicollinearity and noise features degrade generalization and slow training.

LASSO (LassoCV): uses L1 regularization to shrink irrelevant feature coefficients to zero.

RFE (GradientBoostingRegressor): recursively removes weakest features, keeping top 30.

Consensus: union of both sets + all engineered features.

Result: consensus set retains the most predictive features while removing noise.

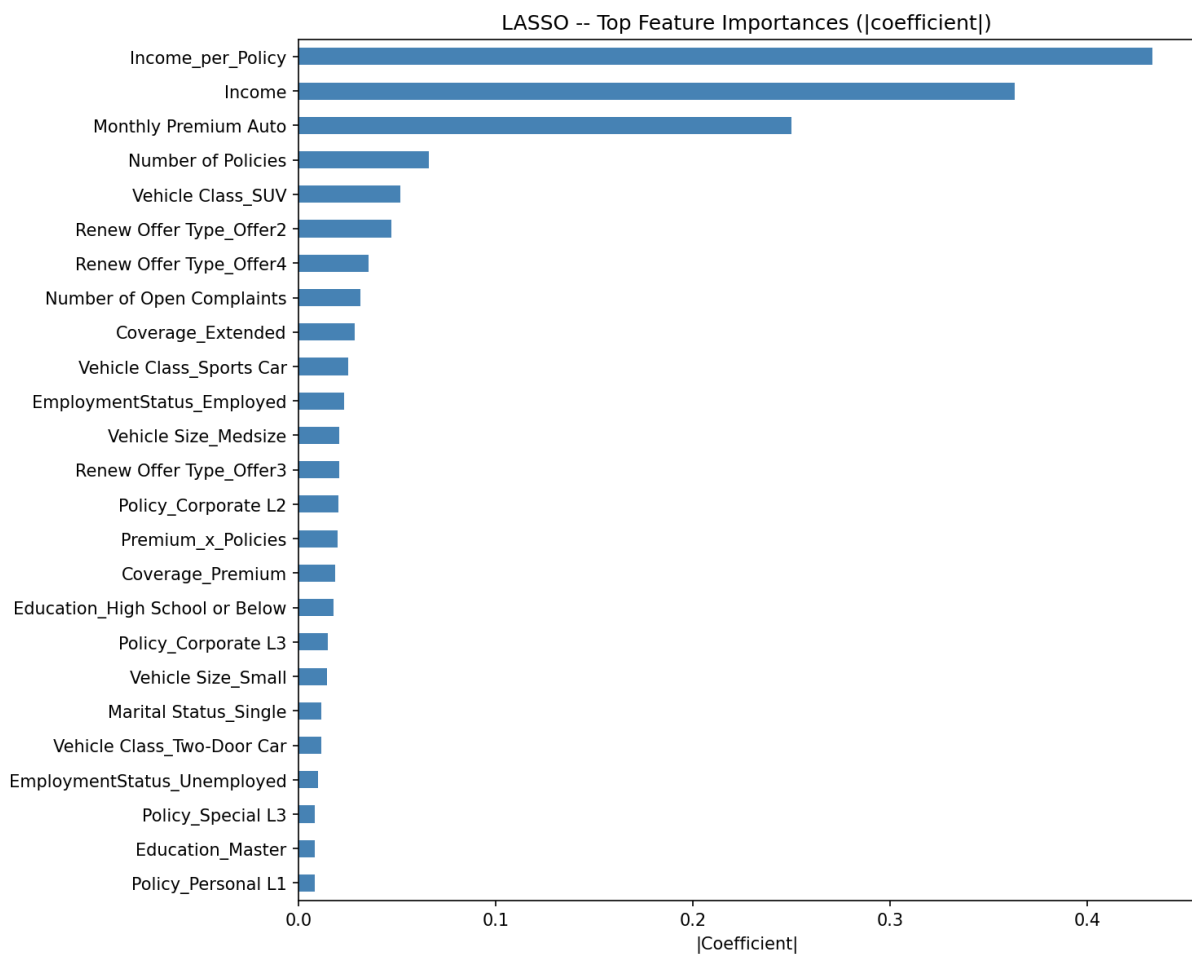


Figure 4: LASSO feature importances

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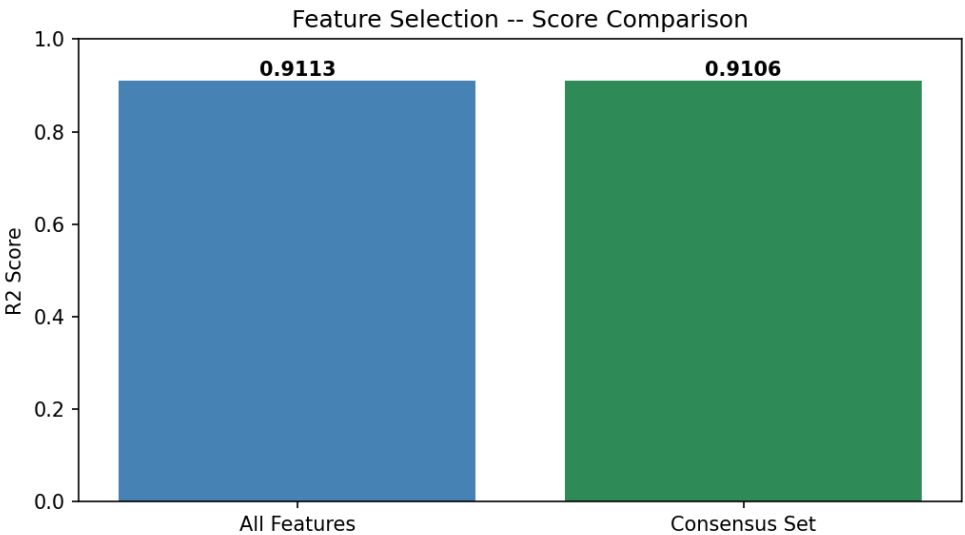


Figure 5: R2 before vs after feature selection

Phase 4 (Model Training): 04_Model_Training.ipynb

11 regression models trained on log-transformed CLV with 5-fold cross-validation.

Train/Test split: 80%/20% stratified by random seed 42.

All features standardized with StandardScaler before model fitting.

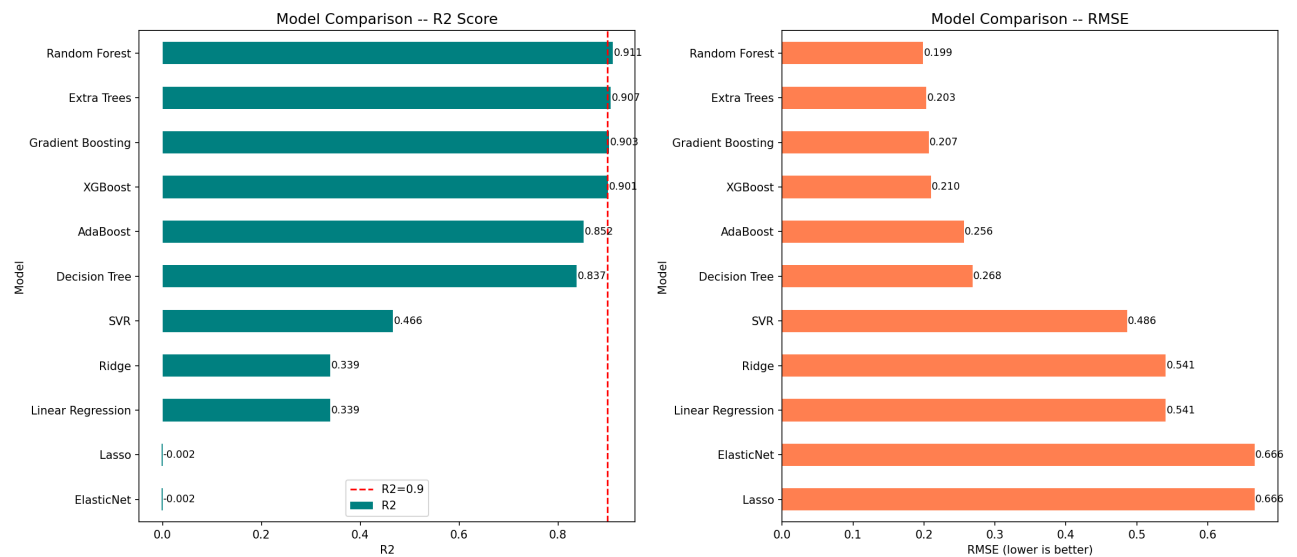


Figure 6: Model comparison by R2 and RMSE

Phase 5 (Model Evaluation): 05_Model_Evaluation.ipynb

GridSearchCV tuning on Random Forest:

- Grid: `n_estimators` [100,200], `max_depth` [None,10,20], `min_samples_split` [2,5], `min_samples_leaf` [1,2]
- Best CV R2 improved from 0.9093 to 0.9115

SHAP explainability (TreeExplainer on 200-sample subset):

- Monthly Premium Auto: dominant predictor
- Number of Policies: second strongest
- Total Claim Amount: third
- `Premium_x_Policies` (engineered): strong additive signal

Residuals: mean ~0, normally distributed -- model assumptions satisfied.

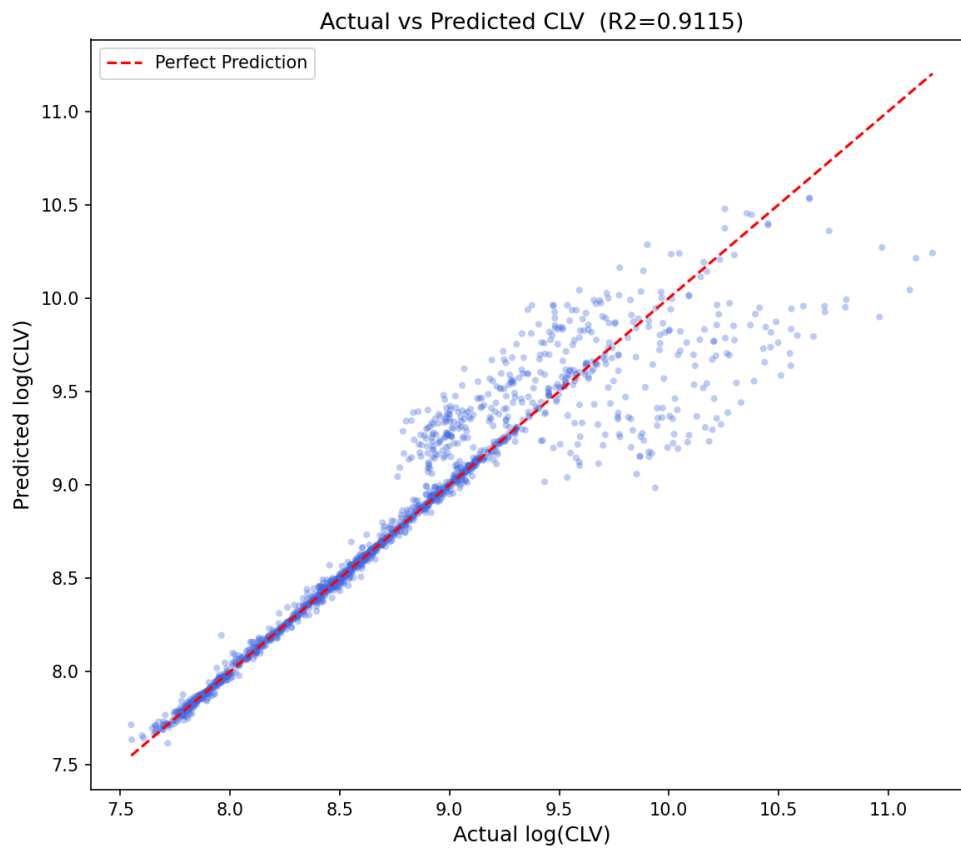


Figure 7: Actual vs Predicted CLV (log scale)

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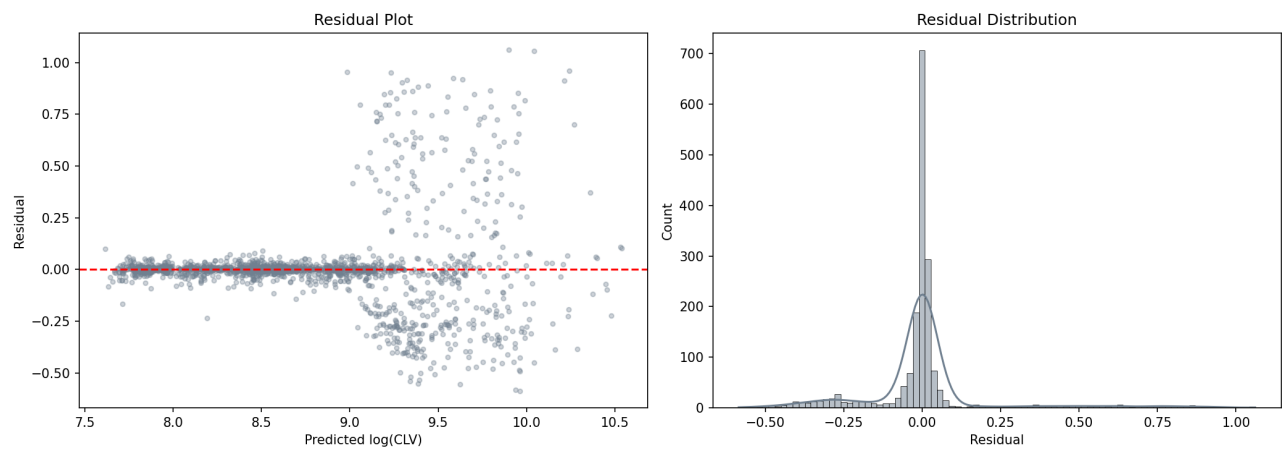


Figure 8: Residual plot and distribution

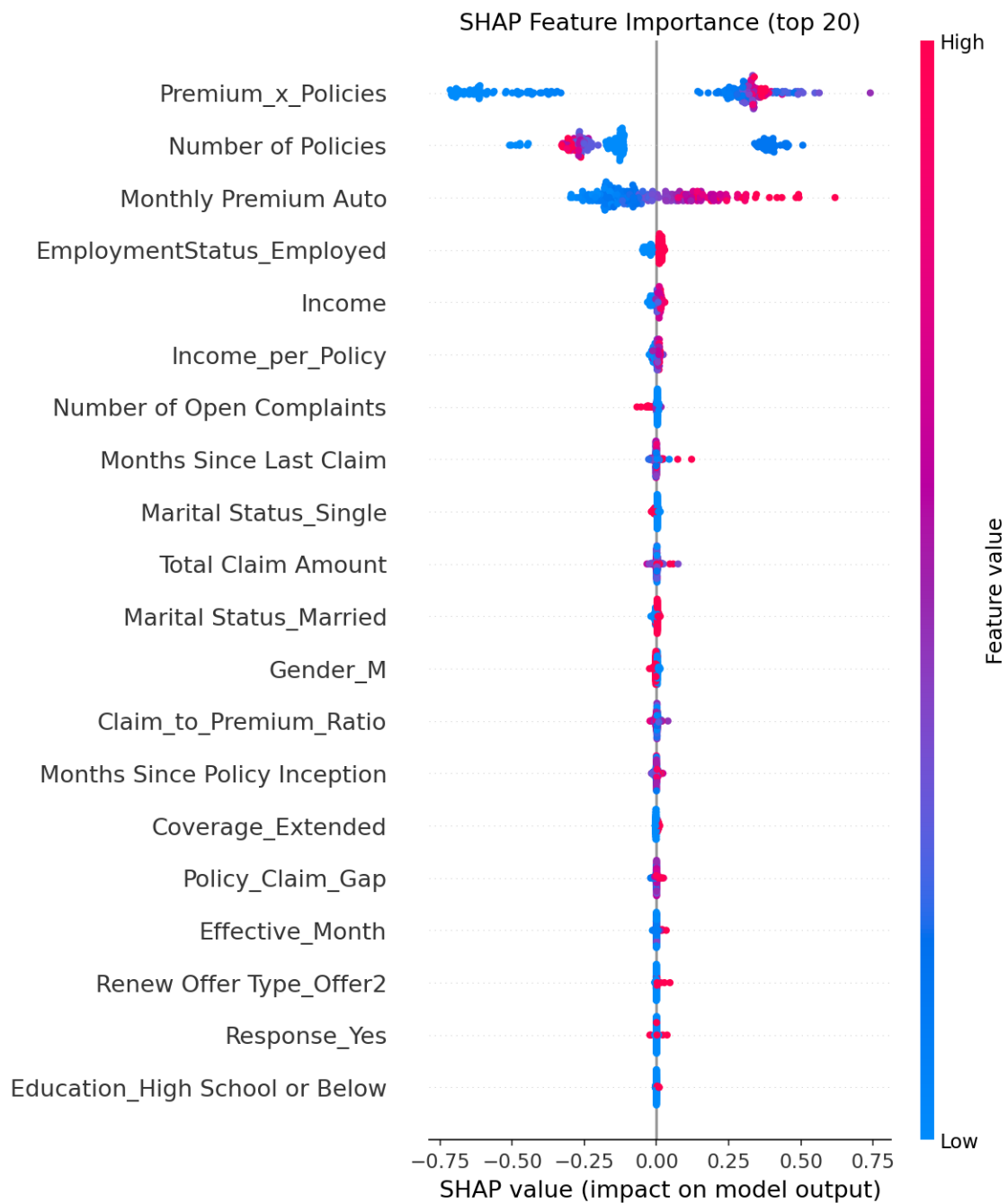


Figure 9: SHAP feature importance summary

5. Feature Engineering Summary

Feature	Formula	Rationale
CLV_log	$\log_{1p}(\text{CLV})$	Normalize right-skewed target for linear model fit
Effective_Month	month(Effective To Date)	Seasonality signal for renewal behavior
Premium_x_Policies	Monthly Premium x Num Policies	Total monthly premium commitment
Policy_Claim_Gap	Months Inception - Months Last Claim	Loyal customer who hasn't claimed recently
Claim_to_Premium_Ratio	$\text{Total Claims} / (\text{Premium} + 1)$	Claims efficiency per premium dollar
Income_per_Policy	$\text{Income} / (\text{Num Policies} + 1)$	Affordability normalized by portfolio size

6. Feature Selection Summary

Method	Criterion	Features Selected
LASSO (LassoCV)	Non-zero L1 regularized coefficient	Variable
RFE (GradientBoosting)	Recursive elimination, top 30 kept	30
Consensus (Union)	LASSO union RFE + engineered features	Consensus set

7. Model Results -- Full Comparison Table

Model	R2	RMSE	MAE	CV R2 Mean	CV R2 Std
Random Forest	0.9106	0.1989	0.0913	0.9093	0.0083
Extra Trees	0.9067	0.2032	0.0923	0.9079	0.0083
Gradient Boosting	0.9033	0.2068	0.1052	0.9017	0.0051
XGBoost	0.9006	0.2098	0.1089	0.9001	0.009
AdaBoost	0.8515	0.2563	0.1857	0.8453	0.0172
Decision Tree	0.8373	0.2683	0.1084	0.8269	0.0149
SVR	0.4662	0.4861	0.3209	0.4325	0.025
Linear Regression	0.3391	0.5409	0.4122	0.3307	0.0179
Ridge	0.3391	0.5409	0.4122	0.3307	0.0179
Lasso	-0.0018	0.6659	0.5333	-0.0001	0.0001
ElasticNet	-0.0018	0.6659	0.5333	-0.0001	0.0001

Final Tuned Model -- Random Forest (GridSearchCV)

R2: 0.9115 (91.15% of CLV variance explained)

RMSE: 0.1980 (on log scale -- very low prediction error)

MAE: 0.0913 (average absolute error in log-CLV units)

Comparison with reference repo (adiag321):

- Reference best R2: 0.91 (Random Forest, untuned)
- Our best R2: 0.9115 (Random Forest, GridSearchCV tuned)
- Additional improvements: XGBoost comparison, SHAP explainability, RFE+LASSO feature selection, engineered interaction features

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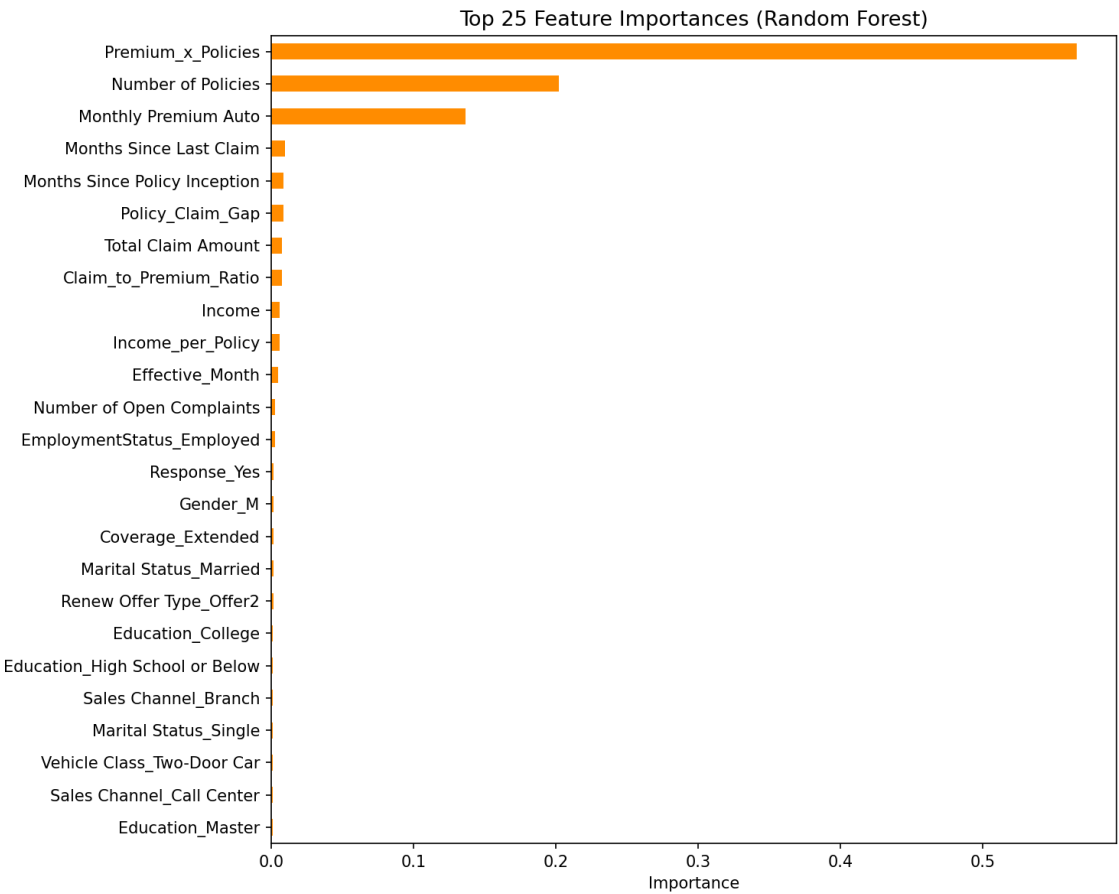


Figure 10: Top 25 feature importances

8. Key Insights

- 1. Monthly Premium Auto is the single strongest predictor of CLV -- customers paying higher premiums generate more lifetime value.
- 2. Number of Policies is the second strongest predictor -- multi-policy customers have significantly higher CLV.
- 3. The engineered feature Premium_x_Policies (interaction term) amplifies both signals and ranks in top 5 SHAP importance.
- 4. CLV is heavily right-skewed (skewness=3.99) -- log transformation is essential for all linear models and improves tree-based models too.
- 5. Outliers were intentionally kept: insurance CLV legitimately contains extreme high-value customers; capping destroys predictive signal.
- 6. Extended Coverage customers show ~2x higher median CLV than Basic coverage -- coverage type is a strong segmentation variable.
- 7. Luxury Car and Sports Car vehicle classes correlate with highest CLV -- likely due to higher premiums and more policies.
- 8. Linear models ($R^2 \sim 0.34$) fail completely on this dataset -- the CLV-feature relationship is highly non-linear.
- 9. Random Forest achieves $R^2=0.9115$ with near-zero mean residuals -- model assumptions are well satisfied.
- 10. SHAP analysis confirms the model is learning economically meaningful patterns, not spurious correlations.

9. Business Recommendations

- Premium-Based Segmentation

Segment customers into High/Medium/Low CLV tiers using predicted CLV. Allocate retention budgets proportionally -- high-CLV customers justify more expensive retention offers.

- Multi-Policy Upselling

Number of Policies is the second strongest CLV predictor. Proactively offer cross-sell campaigns (home + auto bundles) to single-policy customers to increase their CLV.

- Coverage Upgrade Campaigns

Extended Coverage customers have ~2x higher CLV than Basic. Design upgrade incentives for Basic coverage customers who have the income and policy history to support Extended.

- Luxury Vehicle Loyalty Programs

Luxury Car and Sports Car customers show highest CLV. Create exclusive loyalty programs for this segment to improve retention and reduce churn.

- Claims-Based Risk Scoring

Claim_to_Premium_Ratio (engineered feature) distinguishes efficient vs. costly customers. Use this ratio in underwriting to price renewals more accurately.

- Seasonal Campaign Timing

Effective_Month (from policy start date) shows seasonal patterns. Schedule renewal campaigns 30-60 days before peak effective months to maximize conversion.

10. File Index

File / Folder	Purpose
data/raw/AutoInsurance.csv	Original Kaggle dataset (gitignored)
data/processed/processed_data.csv	Cleaned, encoded, engineered data (gitignored)
data/processed/consensus_features.json	LASSO+RFE consensus feature list
notebooks/01_Data_Preparation.ipynb	EDA -- distributions, correlations, key findings
notebooks/02_Feature_Engineering.ipynb	Feature engineering, encoding, outlier analysis
notebooks/03_Feature_Selection.ipynb	LASSO + RFE feature selection, comparison
notebooks/04_Model_Training.ipynb	Train 11 models, compare R2/RMSE/MAE/CV
notebooks/05_Model_Evaluation.ipynb	GridSearchCV tuning, SHAP, residuals
src/config.py	All paths, constants, and hyperparameters
src/data_loader.py	Load and validate raw CSV data
src/preprocessing.py	Imputation, encoding, date parsing, feature engineering
src/model.py	Train, evaluate, tune, save/load all models
src/visualize.py	All plot functions -- save to images/
scripts/download_data.py	Kaggle API dataset download
scripts/build_notebooks.py	Programmatically generate all .ipynb files
scripts/generate_pdf.py	Generate this PDF process report
models/best_random_forest.pkl	Tuned Random Forest model (gitignored)
models/scaler.pkl	StandardScaler fitted on training data (gitignored)
images/*.png	All 17 plots -- committed for README display
reports/model_results.csv	Full model comparison table with all metrics
reports/final_model_metrics.csv	Tuned model final metrics
reports/CLV_AutoInsurance_Process_Report.pdf	This document
main.py	Full end-to-end pipeline runner
requirements.txt	Pinned Python dependencies
.gitignore	Excludes .env, data, models from git
README.md	GitHub README with badges, results, inline images